

Autonomous Research / "AI Scientist" Landscape (2025 - mid-2026)

A curated, ranked map of domain-general autonomous-research work: end-to-end discovery systems, hypothesis generation and validation, self-driving labs, agent frameworks and scientific foundation/reasoning models, benchmarks, and the peer-review milestones and critiques that frame the field. Scope: primarily Jan 2025 - Jun 2026 (today is 2026-06-25), with seminal 2023-2024 anchors marked as foundational. Domain-general: ML, biology, chemistry/materials, physics, math, and social science - not just biomedicine.

Method and trust: every arxiv ID, DOI, and date below was either fetched from the source abstract/page or corroborated across multiple independent outlets during six parallel web-research passes. Vendor "first ever" claims (Sakana, Autoscience, Intology) conflict with one another and are flagged claimed-vs-verified. Items that could not be fetched directly are collected in the final Verification Flags section. Maturity tags: demo (works, narrow) / benchmarked (measured on a benchmark or open-sourced) / peer-reviewed (passed real review) / deployed (a product or production use).

A central caution runs through everything: the most rigorous results in this space are negative or deflationary (the Ideation-Execution Gap, the A-Lab reanalysis, the AI Scientist failure-rate audits, Robin's 7.5x-vs-1.75x effect-size gap). Read "Open Problems & Critiques" alongside the headline systems.

Must-Read Shortlist (ranked across all threads)

#	Work	Org	Date	Link / ID	Maturity	Why it tops the list
1	Towards an AI co-scientist	Google (DeepMind/Research)	Feb 2025	arxiv 2502.18864	deployed + peer-reviewed	Largest effort; generate-debate-evolve + Elo tournament; wet-lab-validated AML/fibrosis/AMR leads; Nature 2026 + Gemini for Science product
2	The Virtual Lab	Stanford / CZ Biohub	Jul 2025	Nature s41586-025-09442-9	peer-reviewed	Only agent-discovery system in Nature with wet-lab-validated output (92 SARS-CoV-2 nanobodies; 2 validated)
3	Robin	FutureHouse + Oxford	May 2025	arxiv 2505.13400	peer-reviewed	First to automate every intellectual step of a wet-bio loop; surfaced ripasudil for dry AMD (Nature)
4	The AI Scientist v1 / v2	Sakana AI	Aug 2024 / Apr 2025	2408.06292 / 2504.08066	demo / peer-reviewed (workshop)	Field-defining progenitor; v2 = first fully-AI paper to clear (workshop) peer review
5	POPPER	Stanford (SNAP)	Feb 2025	arxiv 2502.09858	peer-reviewed	The hypothesis-validation engine: agentic sequential falsification with strict Type-I error control
6	AlphaEvolve	Google DeepMind	May 2025	DeepMind blog	deployed	Evolutionary coding agent; first improvement on 4x4 matrix-mult in 56 years; real Google datacenter/chip impact
7	MLE-bench	OpenAI	Oct 2024	arxiv 2410.07095	benchmarked	De-facto industry ML-engineering agent benchmark; reported in model cards
8	RE-Bench	METR	Nov 2024	arxiv 2411.15114	benchmarked	Frontier AI-R&D-automation safety eval; head-to-head vs 61 human experts; cited in lab safety frameworks

#	Work	Org	Date	Link / ID	Maturity	Why it tops the list
9	The Ideation-Execution Gap	Stanford (SALT)	Jun 2025	arxiv 2506.20803	peer-reviewed	Field's key negative result: the LLM idea-novelty edge collapses once ideas are actually executed
10	PaperBench	OpenAI	Apr 2025	arxiv 2504.01848	benchmarked	Gold-standard paper-replication eval (8,316 author-graded subtasks); best agent ~21%
11	AlphaProof + AlphaGeometry 2	Google DeepMind	Jul 2024; Nature 2025	Nature s41586-025-09833-y	peer-reviewed	IMO 2024 silver-medal score in formal (Lean) math; strongest verifiable machine-reasoning result
12	Kosmos	Edison Scientific (FutureHouse)	Nov 2025	arxiv 2511.02824	deployed	Most mature productized end-to-end discovery agent; traceable claims; 7 reported discoveries
13	Aviary	FutureHouse	Dec 2024	arxiv 2412.21154	benchmarked	Agent-environment gym under PaperQA2/Robin/BixBench; open agents match frontier at ~100x lower cost
14	Agents4Science 2025	Stanford + Together AI	Oct 2025	agents4science.stanford.edu	peer-reviewed (venue)	First AI-must-be-first-author + AI-reviewer venue; transparent - only 5 fully-AI papers accepted
15	A-Lab (+ reanalysis)	Berkeley/LBNL + DeepMind	Nov 2023; corrected Jan 2026	Nature s41586-023-06734-w	peer-reviewed (corrected/disputed)	Landmark autonomous materials lab AND the field's defining rigor controversy
16	Beel et al. - Evaluating Sakana's AI Scientist	Beel, Kan, Baumgart	Feb 2025	arxiv 2502.14297	peer-reviewed	Definitive empirical takedown: 42% of experiments failed; poor novelty assessment; ~\$6-15/paper

If you read only four systems: AI co-scientist (largest effort, strongest validation), Virtual Lab or Robin (the wet-lab-validated discoveries), AI Scientist v2 (the reference point + the peer-review milestone), and POPPER (the validation rigor the generators lack). Then skim PaperBench / MLE-bench to see how the field is actually scored, and the critiques in Section 6 to calibrate the hype.

1. End-to-End Autonomous Discovery Systems

Systems that aim to run the full loop - generate ideas, run/propose experiments, analyze, write up. Ranked by verified impact and rigor. Architectural arc of the field: single-pipeline (AI Scientist v1) -> role-based teams (Agent Laboratory's PhD/Postdoc/Professor) -> specialist-agent networks (co-scientist's 6 agents + Supervisor). The frontier divide is open-loop (hypothesis-only - co-scientist) vs closed-loop (acts on real results - Robin's wet lab, AI Scientist's computational loop).

1. **Towards an AI co-scientist** - Google (DeepMind/Research/Cloud; Gottweis, Natarajan et al.). Feb 2025, arxiv 2502.18864; Nature May 2026, DOI 10.1038/s41586-026-10644-y (PubMed 42156544). Gemini-2.0 multi-agent generate-debate-evolve-rank ("tournament", online Elo via simulated debate); mostly hypothesis-stage with collaborator wet-lab validation. *Why it matters*: biggest-org entry, in-vitro-validated AML drug-repurposing leads at clinically relevant concentrations, plus a deployed product (Gemini for Science / Trusted Tester). **Maturity: deployed + peer-reviewed.**

2. **The Virtual Lab** - Stanford / CZ Biohub (Swanson, Wu, Bulaong, Pak, Zou). bioRxiv Nov 2024; Nature Jul 2025 (646:716-723), DOI 10.1038/s41586-025-09442-9. A human-guided AI principal-investigator agent convenes specialist + critic agents that ran an ESM + AlphaFold-Multimer + Rosetta pipeline. *Why it matters*: designed 92 novel SARS-CoV-2 nanobody binders (two experimentally validated vs JN.1) - published in the highest-rigor venue here. **Maturity: peer-reviewed (Nature)**.
3. **Robin** - FutureHouse + Oxford (Ghareeb, Chang, Mitchener, Yiu, White, Rodriques et al.). May 2025, arxiv 2505.13400; later Nature DOI 10.1038/s41586-026-10652-y. Orchestrates three FutureHouse agents (Crow/Falcon literature, Finch data-analysis) into one closed loop with BTL tournament ranking. *Why it matters*: claims first system to automate ALL intellectual steps of a wet-bio loop, surfacing ripasudil as a novel dry-AMD candidate; humans ran only the bench. Already has a full guide in this repo (robin-guide.md). See the "verification paradox" critique in Section 6. **Maturity: peer-reviewed**.
4. **The AI Scientist (v1)** - Sakana AI (C. Lu, Lange, Foerster, Clune, Ha). Aug 2024, arxiv 2408.06292. First end-to-end "idea -> Aider code -> experiment -> paper -> automated review" pipeline (template-bound, ML-only, ~\$15/paper). *Why it matters*: the most-cited progenitor that launched the entire genre. **Maturity: demo**.
5. **The AI Scientist-v2** - Sakana AI (Yamada, Lange, C. Lu, Hu, Foerster, Clune, Ha). Apr 2025, arxiv 2504.08066. Template-free agentic tree-search; idea -> experiments -> full paper, with VLM-reviewed figures. *Why it matters*: produced the first fully-AI-generated manuscript to pass peer review (ICLR 2025 ICBINB workshop, score 6.33), then withdrawn by prior agreement. **Maturity: peer-reviewed (workshop)**.
6. **Kosmos** - Edison Scientific (FutureHouse spinout; Mitchener, Yiu, Chang +34). Nov 2025, arxiv 2511.02824. A 12-hour data-driven discovery agent with a world-model shared across data-analysis and literature agents; ~1,500 papers and ~42k lines of code per run, every claim traceable to code + citation. *Why it matters*: strongest commercial deployment (\$200/run platform), with 7 reported discoveries. **Maturity: deployed**.
7. **AI-Researcher** - HKUDS / HKU (Tang, Xia, Z. Li, C. Huang). May 2025, arxiv 2505.18705. Fully-automated pipeline (lit review -> ideation -> implementation -> validation -> manuscript) shipping Scientist-Bench. *Why it matters*: best-validated open-source "AI co-scientist" alternative; NeurIPS 2025 + productized (novix.science). **Maturity: peer-reviewed + deployed**.
8. **Agent Laboratory** - AMD + Johns Hopkins (Schmidgall, Su, Z. Wang, Moor et al.). Jan 2025, arxiv 2501.04227. Human-seeded pipeline with PhD/postdoc/ML-engineer/professor agents across lit review, experimentation, report writing. *Why it matters*: widely-used open framework with honest human-in-the-loop framing and ~84% cost reduction vs prior auto-research. **Maturity: benchmarked / open-source**.
9. **data-to-paper** - Technion (Tal Ifargan, Lukas Hafner, Maor Kern, Ori Alcalay, Roy Kishony). Apr 2024, arxiv 2404.17605; NEJM AI Dec 2024, DOI 10.1056/AIoa2400555. Data -> hypothesis -> code -> full paper with verifiable claim-to-data provenance and a human co-pilot for hard steps. *Why it matters*: an early, rigor-first design centered on traceability rather than raw autonomy. **Maturity: peer-reviewed (NEJM AI)**. (Author correction 2026-07-03: first author is Tal Ifargan, not "Ben-Kish"; verified against arXiv + NEJM AI. See Section 7.)
10. **Dolphin** - Shanghai AI Lab / InternScience (Yuan, Yan, B. Zhang, Ouyang, Qiao, B. Zhou). Jan 2025, arxiv 2501.03916. Early closed-loop "think -> practice -> feedback" auto-research where results feed the next ideas. *Why it matters*: ACL 2025 main-conference accepted; clean early closed-loop design (ML-only). **Maturity: peer-reviewed (ACL main)**.

11. **InternAgent (formerly NovelSeek)** - Shanghai AI Lab / InternScience (25 authors). May 2025, arxiv 2505.16938. Unified closed-loop multi-agent framework spanning 12 scientific tasks/domains. *Why it matters*: broadest multi-domain coverage; v1.5 successor active in 2026. **Maturity: benchmarked.** (Naming note: NovelSeek = InternAgent, renamed.)
 12. **Zochi / Tempest** - Intology AI. Paper Mar 2025, arxiv 2503.10619; claimed ACL 2025 main acceptance. "Tempest" does multi-turn LLM jailbreaking via tree search. *Why it matters*: boldest milestone claim (first AI author at a main A* conference), heavily disputed - see Critiques. **Maturity: peer-reviewed (claimed) / disputed.** Flag: arxiv lists humans Andy Zhou and Ron Arel as authors of record; "Zochi" authorship is vendor framing.
 13. **CycleResearcher** - open-source (Review-5k/Research-14k datasets). Nov 2024, arxiv 2411.00816. Iterative preference-trained LLM (12B/72B/123B) doing research, paired with a CycleReviewer for RL-style feedback. *Why it matters*: open-weight models reaching near-human preprint quality in simulated review only. **Maturity: benchmarked (open weights).**
 14. **Autoscience "Carl"** - Autoscience Institute. Mar 2025, company blog / press (no arxiv). Claimed 3/4 ICLR 2025 workshop acceptances; raised \$14M. *Why it matters*: another conflicting "first peer-reviewed AI paper" claim, weakest documentation (no technical report, venues/scores undisclosed). **Maturity: claim-stage / unverified.**
 15. **EvoScientist** - Yougang Lyu et al. Mar 2026, arxiv 2603.08127. Evolving multi-agent AI-scientist with three roles (Researcher, Engineer, Evolution Manager) for end-to-end discovery. *Why it matters*: recent evolutionary multi-agent design; CONFIRMED real in the 2026-06-25 deep pass (a prior draft mis-flagged it as spurious). **Maturity: demo.**
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2. Autonomous Hypothesis Generation & Validation

Generating research ideas/hypotheses AND (crucially) validating or falsifying them. Ranked by rigor of validation. The natural pipeline: generate -> rank -> POPPER adjudicates with a calibrated false-positive rate -> spend on wet-lab / policy.

1. **POPPER** - Stanford / SNAP (Huang, Y. Jin, R. Li, M. Li, Candès, Leskovec). Feb 2025, arxiv 2502.09858. Validates free-form hypotheses via LLM agents that design and run sequential falsification experiments; e-values / e-processes give any-time-valid, frequentist Type-I error control. *Why it matters*: the reference for the "validate" half of the loop; matches human biologists at ~10x speed across six domains (bio/econ/sociology). It is the only entry here that emits a calibrated error rate on hypotheses it did not generate. **Maturity: peer-reviewed (ICML 2025).**
2. **The Ideation-Execution Gap** - Stanford / SALT (Si, Hashimoto, Yang). Jun 2025, arxiv 2506.20803. 43 experts spent 100+ hours each actually executing LLM-generated vs human ideas, scoring before and after. *Why it matters*: the field's most important negative result - the apparent novelty edge of LLM ideas does not survive execution. **Maturity: peer-reviewed (human study).**
3. **Can LLMs Generate Novel Research Ideas?** - Stanford (Si, Yang, Hashimoto). Sep 2024, arxiv 2409.04109. Largest head-to-head human study (100+ NLP researchers): LLM ideas rated more novel (5.64 vs 4.84) but less feasible. *Why it matters*: defines the field's evaluation paradigm; the companion to the Ideation-Execution Gap above. **Maturity: peer-reviewed (ICLR 2025).** Flag: Sep 2024, just outside the window; foundational.

4. **SciAgents** - MIT (Ghafarollahi, Buehler). Sep 2024, arxiv 2409.05556. Multi-agent generation+refinement over ontological knowledge graphs for materials-science hypotheses. *Why it matters*: influential knowledge-graph-grounded generation. **Maturity: peer-reviewed**. Flag: Sep 2024, foundational.
5. **MOOSE-Chem** - (Z. Yang, Bing, Cambria et al.). Oct 2024, arxiv 2410.07076. Decomposes chemistry hypotheses into background + inspirations and rediscovers real 2024 Nature/Science findings. *Why it matters*: rare ground-truth validation (rediscovers unseen real papers). **Maturity: peer-reviewed (ICLR 2025)**.
6. **MOOSE-Chem2** - (Z. Yang, W. Liu, Gao, Bing, Cambria, Zhou et al.). May 2025, arxiv 2505.19209. Fine-grained, experimentally-actionable hypothesis discovery via hierarchical search. *Why it matters*: pushes from coarse ideas to testable methodological detail. **Maturity: benchmarked**.
7. **AI Idea Bench 2025** - (Qiu, H. Zhang, Xu, M. Li, Song, Z. Wang, K. Zhang). Apr 2025, arxiv 2504.14191. Benchmark of 3,495 AI papers + inspired works with leakage-aware tasks (IMCQ/I2I/I2T). *Why it matters*: the best leakage-controlled, ground-truth idea-generation benchmark for the AI domain. **Maturity: benchmarked**. (Distinct from IdeaBench - do not conflate.)
8. **ResearchTown** - UIUC (H. Yu, Hong, Cheng, K. Zhu, You et al.). Dec 2024 (v2 Jun 2025), arxiv 2412.17767. Simulates a research community as an agent-data graph with TextGNN message-passing; ships ResearchBench. *Why it matters*: novel community-simulation framing. **Maturity: peer-reviewed (ICML 2025)**.
9. **IdeaBench** - UVA (S. Guo, Shariatmadari, Xiong, Bekiranov, A. Zhang). Oct 2024, arxiv 2411.02429. Benchmark/dataset profiling LLMs as domain researchers, grounded in influential papers. *Why it matters*: early standardized idea-gen benchmark. **Maturity: peer-reviewed (KDD 2025)**.
10. **Nova** - (X. Hu, Fu et al.). Oct 2024, arxiv 2410.14255. Iterative planning + external-knowledge search to boost idea novelty/diversity. *Why it matters*: strong generation-side diversity method. **Maturity: benchmarked**. Flag: Oct 2024.
11. **SciMON** - (Q. Wang, Downey, Ji, Hope). May 2023, arxiv 2305.14259. Generation framework explicitly optimizing novelty vs prior literature. *Why it matters*: lineage anchor that defined "novelty-bounded" generation. **Maturity: peer-reviewed (ACL 2024)**. Flag: 2023, far outside window; included as lineage.

Note on ranking machinery: Robin uses batch Bradley-Terry-Luce MLE over a fixed comparison set (single-turn judgments); the AI co-scientist uses online sequential Elo with multi-turn debate as the comparison operator. Both share the logistic $\sigma(\theta_i - \theta_j)$ kernel - the same family as RLHF reward models and DPO. Neither emits a calibrated false-positive rate; that gap is exactly what POPPER fills.

3. Autonomous Experimentation / Self-Driving Labs

Physical and robotics/cloud-lab-in-the-loop discovery across chemistry, materials, biology, physics - the physical counterpart to Robin's wet lab. Seminal 2023-2024 systems marked foundational.

1. **Robin** - see Section 1. First peer-reviewed end-to-end loop yielding a real drug candidate, and (per critics) the field's most instructive failure case in autonomous data analysis.
2. **Google AI Co-Scientist** - see Section 1. Multiple collaborator wet-lab validations; the superbug/AMR result (below) is its most-publicized and most-contested.

3. **Ginkgo Cloud Lab + OpenAI** - Ginkgo Bioworks / OpenAI. Cloud Lab launch ~Mar 2026; GPT-5 closed-loop result late 2025 (PRNewswire). Web-accessible autonomous bio-lab (70+ instruments); GPT-5 ran ~36,000 conditions over 6 cycles, cutting cell-free protein-synthesis cost ~40%. *Why it matters*: the most concrete deployed closed-loop result at scale on real bio infrastructure. **Maturity: deployed**. Flag: details from press releases, not a peer-reviewed paper.
4. **Self-driving labs review (technology + policy)** - Tobias & Wahab, R. Soc. Open Sci. 12(7):250646, Jul 2025, DOI 10.1098/rsos.250646. Surveys SDLs across chem/materials/bio and proposes an SAE-style L0-L5 autonomy scale. *Why it matters*: the framing everyone now cites; sober claim that most "self-driving labs" are L2, a few L3, L5 aspirational. **Maturity: peer-reviewed (survey)**.
5. **A-Lab** (*foundational; resolution in-window*) - Berkeley/LBNL (Szymanski, Zeng, Ceder, Persson) + DeepMind. Nature Nov 2023, DOI 10.1038/s41586-023-06734-w; **correction Jan 19 2026**. Autonomous inorganic-materials lab claiming 41 of 58 novel syntheses in 17 days. *Why it matters*: seminal materials SDL and the defining rigor controversy (Section 6). **Maturity: peer-reviewed / corrected / disputed**.
6. **Coscientist** (*foundational*) - CMU (Boiko, MacKnight, Kline, Gomes). Nature Dec 2023 (624:570-578), DOI 10.1038/s41586-023-06792-0. GPT-4 planner agent driving cloud-lab robotics to execute organic reactions (Pd cross-coupling optimization). *Why it matters*: the paper that launched the LLM-agent-runs-a-lab paradigm; lineage root for much of this thread. **Maturity: peer-reviewed**.
7. **GNoME** (*foundational*) - Google DeepMind (Merchant, Cubuk). Nature Nov 2023, DOI 10.1038/s41586-023-06735-9. Graph-network high-throughput discovery of ~381k stable inorganic crystals (~736 later synthesized/confirmed). *Why it matters*: massively expanded the Materials Project, but faces 2025 novelty/duplicate disputes (contested). **Maturity: peer-reviewed (contested)**.
8. **Liverpool two-robot autonomous chemist** - Cooper group, Liverpool. Nature Nov 2024, DOI 10.1038/s41586-024-08173-7. Two mobile robots cooperatively run exploratory synthesis with Chemspeed + UPLC-MS + NMR and AI decisions. *Why it matters*: pushes from optimization to open-ended exploratory chemistry; Google-funded CO2-capture "hive mind" follow-on. **Maturity: peer-reviewed**.
9. **ORGANA** - Toronto/Vector (Darvish, Skreta, Aspuru-Guzik, Garg, Shkurti). Matter Nov 2024, arxiv 2401.06949. LLM robotic assistant: natural language -> chemical description language -> multi-step chemistry with vision feedback and scheduling on ~2/3 off-the-shelf hardware. *Why it matters*: strong reproducible embodied-robot system. **Maturity: peer-reviewed**.
10. **Lila Sciences** - Flagship Pioneering spinout. Stealth exit Mar 2025; ~\$550M raised, >\$1.3B valuation. "Scientific superintelligence" pairing foundation models with autonomous wet+dry labs. *Why it matters*: best-funded pure-play autonomous-lab startup; a bellwether - but no flagship validated discovery yet. **Maturity: deployed (early)**. Flag: company claims, not peer-reviewed.
11. **Periodic Labs** - Fedus (ex-OpenAI) + Cubuk (ex-DeepMind/GNoME). Sep 2025; a16z-led \$300M seed. Autonomous labs + large models for physical sciences, starting with superconductors. *Why it matters*: record seed and elite pedigree; too new to judge output. **Maturity: deployed (early/stealth)**. Flag: press-stage.
12. **El Agente (Q)** - Toronto (Zou, Cheng, Aspuru-Guzik et al.). arxiv May 2025, arxiv 2505.02484; Matter Jul 2025. LLM multi-agent that turns natural language into executable quantum-chemistry workflows with hierarchical memory and in-situ debugging. *Why it matters*: brings the autonomous-agent pattern to computational (dry-lab) science; >87% task success. **Maturity: peer-reviewed (dry lab)**.
13. **Argonne Polybot** - Argonne CNM (Jie Xu et al.). 2025. Robot-driven SDL for electronic polymer thin films, co-optimizing conductivity and coating defects. *Why it matters*: DOE-scale deployed materials SDL with concrete device wins. **Maturity: deployed / peer-reviewed**.

14. **MIT CRES**t - MIT (Ju Li group). Sep 2025 (MIT News + ChemRxiv). Multimodal AI copilot: NL goals -> propose/run experiments via robotics + active learning, with vision-language monitoring. *Why it matters*: notable multimodal closed loop; reported record formate fuel-cell performance. **Maturity: demo / preprint**. Flag: peer-review status and performance claim unconfirmed.
15. **Emerald Cloud Lab + agents** (*enabling infra*) - Emerald Cloud Lab; CMU Cloud Lab. AI-native remote bio/chem lab with Symbolic Lab Language; the robotic backend behind Coscientist. *Why it matters*: the execution substrate many "autonomous" agents actually run on. **Maturity: deployed (infra)**.

4. Agent Frameworks / Infrastructure & Scientific Foundation/Reasoning Models

The substrate (environments, tool-using agents, preprint-sharing) and the discovery-capable models. All arxiv IDs/DOIs below were source-verified except where flagged.

Frameworks & infrastructure

1. **Aviary** - FutureHouse + Rochester + Francis Crick Inst. (Narayanan, Braza, Griffiths, White, Rodriques). Dec 2024, arxiv 2412.21154. Open gymnasium framing language agents as policies over language decision processes; 3 scientific envs (molecular cloning, lit-QA, protein stability). *Why it matters*: open non-frontier agents match/beat frontier agents and human experts at up to ~100x lower inference cost; substrate under PaperQA2/Robin/BixBench. **Maturity: benchmarked (open-source)**.
2. **PaperQA2** - FutureHouse (Skarlinski, Cox, Laurent, Braza, Rodriques, White). Sep 2024, arxiv 2409.13740. Agentic RAG for retrieval/summarization/contradiction-detection over scientific literature. *Why it matters*: first agent to beat PhD/postdoc biologists on literature search (LitQA2); the "read the field" primitive powering the FutureHouse platform. **Maturity: deployed**.
3. **AgentRxiv** - Schmidgall & Moor. Mar 2025, arxiv 2503.18102. A shared "preprint server" letting agent labs upload, retrieve, and build on each other's research. *Why it matters*: collaborating agent labs progress faster (+13.7% relative on MATH-500) than isolated ones; first infra for cumulative autonomous research. **Maturity: demo (open-source)**.
4. **WikiCrow / FutureHouse Platform** - FutureHouse. Sep 2024 / May 2025. WikiCrow auto-generates Wikipedia-style articles for all ~20,000 human protein-coding genes (rated more accurate than human Wikipedia on average); the public platform exposes Crow (quick QA), Falcon (deep review), Owl (has-anyone-done-X), Phoenix (chemistry, ChemCrow-based). *Why it matters*: first public superhuman scientific-search agents via API. **Maturity: deployed**.
5. **ChemCrow** (*foundational*) - EPFL/Rochester (Bran, Cox, White, Schwaller). Apr 2023, arxiv 2304.05376; Nature Mach. Intell. May 2024. GPT-4 chemistry agent with 18 expert tools that planned and physically executed real syntheses. *Why it matters*: the template the tool-augmented-agent field builds on; basis for FutureHouse Phoenix. **Maturity: peer-reviewed**.
6. **Curie** - Univ. Michigan + Cisco Research (Kon, J. Liu, Chowdhury, A. Chen). Feb 2025, arxiv 2502.16069. Agent framework for rigorous, reliable experimentation, with a 46-question CS eval (3.4x over the strongest baseline). *Why it matters*: explicit focus on experimental rigor. **Maturity: benchmarked (framework)**. (Note: a separate Google benchmark named CURIE, arxiv 2503.13517, measures long-context scientific reasoning - do not conflate.)

7. **Frontier-lab science programs (deployed)** - Anthropic "AI for Science" (May 2025, API credits for bio/life-science research); Anthropic "Claude for Life Sciences" (Oct 2025, Sonnet 4.5 + connectors/Agent Skills; 0.83 vs 0.79 human on Protocol QA); OpenAI "for Science" (Oct 2025, Kevin Weil). *Why it matters:* the major-lab push to operationalize agents for working scientists. **Maturity: deployed.** Flag: OpenAI for Science page returned 403; corroborated via secondary coverage.

Scientific foundation & reasoning models

1. **AlphaEvolve** - Google DeepMind. May 2025, DeepMind blog/whitepaper (no arxiv). Gemini-powered evolutionary coding agent that evolves whole codebases via LLM + automated evaluators. *Why it matters:* found a 48-multiplication 4x4 complex matrix-mult scheme (beating Strassen's 49, first improvement in 56 years), improved ~20% of 50 open math problems, and recovered ~0.7% of Google datacenter compute. **Maturity: deployed.** Flag: blog/whitepaper only; narrower than full "AI Scientist" scope.
2. **AlphaProof** - Google DeepMind. IMO Jul 2024; Nature Nov 2025, DOI 10.1038/s41586-025-09833-y ("Olympiad-level formal mathematical reasoning with RL"). AlphaZero-style RL agent that proves theorems in formal Lean with test-time RL. *Why it matters:* IMO 2024 silver level (28/42; solved the hardest problem) with verifiable, hallucination-free proofs. **Maturity: peer-reviewed.**
3. **AlphaGeometry 2** - Google DeepMind (Chervonyi, Trinh, Luong). Feb 2025, arxiv 2502.03544. Neuro-symbolic olympiad-geometry solver (Gemini LM + knowledge-sharing search). *Why it matters:* gold-medalist geometry (solves 84% of last-25-years IMO geometry); the geometry half of the IMO-2024 silver system. **Maturity: benchmarked.**
4. **Gemini Deep Think (IMO 2025)** - Google DeepMind. Jul 2025, DeepMind blog. General-purpose LLM, natural-language proofs, no tools. *Why it matters:* first officially certified AI IMO gold (35/42, 5/6); an OpenAI experimental model also hit 35/42 (uncertified). **Maturity: deployed/announced.**
5. **Evo 2** - Arc Institute + NVIDIA + Stanford. bioRxiv Feb 2025 (2025.02.18.638918); Nature (10.1038/s41586-026-10176-5, online 2026-03-04). Genomic foundation model (40B flagship + 7B/20B/1B variants, >9T nucleotides, 1-megabase context, all domains of life). *Why it matters:* zero-shot variant-effect prediction (e.g. BRCA1) + genome-scale DNA generation; largest open biology FM. **Maturity: peer-reviewed (Nature).** Verified: Nature DOI + param counts CONFIRMED (deep pass 2026-06-25).
6. **ESM3** - EvolutionaryScale (Hayes et al.). bioRxiv Jul 2024; Science Jan 2025, DOI 10.1126/science.ads0018. Multimodal generative protein LM (up to 98B params) over sequence/structure/function. *Why it matters:* generated a novel fluorescent protein (esmGFP) ~500M years of evolutionary distance from known GFPs. **Maturity: peer-reviewed.**
7. **AlphaFold 3** - Google DeepMind + Isomorphic Labs (Abramson et al.). Nature May 2024, DOI 10.1038/s41586-024-07487-w; weights to academia Nov 2024. Diffusion predictor of structure + interactions for proteins, nucleic acids, ligands, ions. *Why it matters:* extended folding to all biomolecular complexes; free AlphaFold Server for non-commercial use - the structural-biology workhorse. **Maturity: deployed.**
8. **Boltz-2** - MIT Jameel Clinic + Recursion. bioRxiv Jun 2025. Open (MIT-license) biomolecular FM jointly predicting structure + binding affinity. *Why it matters:* first DL model approaching FEP-physics affinity accuracy ~1000x faster; a fully open AF3-class alternative (builds on Boltz-1, Nov 2024). **Maturity: benchmarked (open).**
9. **MatterGen** - Microsoft Research AI for Science. Nature Jan 2025, DOI 10.1038/s41586-025-08628-5. Diffusion generative model for novel inorganic crystals conditioned on property constraints. *Why it matters:* structures ~2x more likely novel+stable, ~10x closer to the energy minimum; inverse materials design. **Maturity: peer-reviewed.**

10. **Aurora** - Microsoft Research (Bodnar et al.). arxiv May 2024 (2405.13063); Nature May 2025, DOI 10.1038/s41586-025-09005-y. 1.3B-param Earth-system foundation model (weather, air quality, ocean waves, cyclone tracks). *Why it matters*: beats operational IFS/GraphCast on >91% of targets at orders-of-magnitude lower cost; first large atmosphere FM. **Maturity**: peer-reviewed.
 11. **Frontier reasoning on science benchmarks** - OpenAI / Google / Anthropic, 2025-2026. GPT-5.x, Gemini Deep Think, Claude on FrontierMath / Humanity's Last Exam / GPQA. *Why it matters*: FrontierMath now ~40-48% (GPT-5.x), GPQA Diamond >93%, but HLE still well below ~35% (large headroom). **Maturity**: deployed. Flag: scores shift fast and many are from search snippets, not page-verified.
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5. Benchmarks & Evaluations

How the field measures autonomous-research and ML-engineering agents. All arxiv IDs fetched/verified. Ranked by adoption and rigor. Picking one: MLE-bench / RE-Bench for ML-engineering agents; PaperBench (replication) vs CORE-Bench (reproduction); ScienceAgentBench / DiscoveryBench / BixBench for data-driven scientific reasoning.

1. **MLE-bench** - OpenAI. Oct 2024, arxiv 2410.07095. 75 real Kaggle ML-engineering competitions, human-leaderboard-anchored. Headline: o1-preview+AIDE reaches >=bronze in only 16.9%. *Maturity*: widely-adopted.
2. **RE-Bench** - METR. Nov 2024, arxiv 2411.15114. 7 open-ended frontier ML-R&D environments vs 61 human experts. Headline: agents beat humans ~4x at a 2h budget, but humans win at 8h+. *Maturity*: widely-adopted (safety policy).
3. **PaperBench** - OpenAI. Apr 2025, arxiv 2504.01848. Replicate 20 ICML 2024 papers from scratch; 8,316 author-co-developed rubric subtasks, LLM-judge graded. Headline: best agent (Claude 3.5 Sonnet) ~21%. *Maturity*: widely-adopted.
4. **AstaBench** - Allen Institute for AI (Bragg, D'Arcy, Balepur +35). Oct 2025, arxiv 2510.21652. 2,400+ problems across domains/discovery stages; 57 agents evaluated. Conclusion: "AI remains far from solving science research assistance." *Maturity*: released (likely 2026 standard).
5. **ScienceAgentBench** - Ohio State (Chen, Ning, Su, Sun +20). Oct 2024, arxiv 2410.05080. 102 data-driven discovery tasks from 44 peer-reviewed papers; self-contained Python outputs. Headline: best agent ~32.4% (o1-preview 42.2%). *Maturity*: widely-adopted.
6. **SciCode** - multi-university consortium. Jul 2024, arxiv 2407.13168. 338 subproblems / 80 problems across 16 natural-science subfields. Headline: top model solves only 4.6% in the realistic setting. *Maturity*: widely-adopted (active leaderboard).
7. **DiscoveryBench** - AI2 (Majumder, Surana, Khot, Sabharwal, Clark). Jul 2024, arxiv 2407.01725. Multi-step data-driven discovery: 264 real + 903 synthetic tasks across 6 domains. Headline: best system ~25%. (Used by POPPER.) *Maturity*: widely-cited.
8. **DiscoveryWorld** - AI2 (Jansen, Côté, Khot, Clark). Jun 2024, arxiv 2406.06769. Full hypothesize->experiment->analyze->conclude virtual environment; 120 tasks. *Maturity*: widely-cited (NeurIPS 2024 D&B).
9. **CORE-Bench** - Princeton (Siegel, Kapoor, Nadgir, Stroebel, Narayanan). Sep 2024, arxiv 2409.11363. Computational reproducibility (install -> run -> read): 270 tasks, 90 papers. Headline: best agent ~21% on the hardest level. *Maturity*: adopted.

10. **MLGym / MLGym-Bench** - Meta GenAI/FAIR + UCSB (Nathani, Madaan, Raileanu, Foerster). Feb 2025, arxiv 2502.14499. First Gym/RL environment + 13 open-ended AI-research tasks; enables training agents. Headline: frontier LLMs tune well but generate no novel algorithms. *Maturity: released (growing)*.
 11. **SUPER** - AI2 (Bogin, Yang, Gupta, Richardson, Clark, Sabharwal, Khot). Sep 2024, arxiv 2409.07440. Setting up + executing tasks from real ML/NLP repos. Headline: GPT-4o ~16.3% end-to-end. *Maturity: adopted (EMNLP 2024)*.
 12. **DSBench** - UT Dallas + Tencent AI Lab. Sep 2024, arxiv 2409.07703. 466 data-analysis + 74 data-modeling tasks. Headline: best agent ~34% analysis. *Maturity: adopted (ICLR 2025)*.
 13. **BixBench** - FutureHouse + ScienceMachine (Mitchener, Laurent, Andonian, White, Rodriques). Mar 2025, arxiv 2503.00096. ~50 real bioinformatics scenarios / ~300 open-answer questions (introduces the Finch agent). Headline: frontier models ~17% open-answer (~22% agentic SOTA). *Maturity: gaining adoption*.
 14. **LAB-Bench** - FutureHouse (Laurent, Janizek, White, Rodriques). Jul 2024, arxiv 2407.10362. 2,400+ MCQs on practical biology-research skills vs expert biologists. *Maturity: widely-adopted*.
 15. **AAAR-1.0** - Penn State + UC Davis (Lou, Kamoi, W. Yin, L. Huang). Oct 2024, arxiv 2410.22394. Expert research tasks: equation inference, experiment design, paper weakness, review critique. *Maturity: adopted (ICML 2025)*.
 16. **The Automated LLM Speedrunning Benchmark** - Meta FAIR (B. Zhao, Magka, M. Jiang + 20). Jun 2025, arxiv 2506.22419. Reproduce 19 known nanoGPT-speedrun improvements given prior script + hints. Headline: SOTA agents struggle even with detailed hints. *Maturity: released*.
 17. **MLR-Bench** - (academic). May 2025, arxiv 2505.19955. 201 open-ended ML-research tasks from major conferences covering the full pipeline. *Maturity: released*.
 18. **MLAgentBench (foundational)** - Stanford (Q. Huang, Vora, Liang, Leskovec). Oct 2023, arxiv 2310.03302. 13 open-ended ML experimentation tasks. *Maturity: widely-cited*. Flag: Oct 2023, pre-window; the foundational ML-agent gym.
 19. **Other coverage** (narrower/newer): **ResearchBench** (2503.21248), **SciReplicate-Bench** (2504.00255), **ResearchArena** (2406.10291), **ResearchGym** (2602.15112, Feb 2026, very new), **CURIE** scientific-reasoning benchmark (2503.13517), and methodology paper **EAIRA** (2502.20309). Note: *IdeaBench* (2411.02429) and *AI Idea Bench 2025* (2504.14191) are two distinct benchmarks. The headline story across all of these: numbers stay low, so autonomous analysis is far from solved.
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6. Peer-Review Milestones & Critiques

(A) Peer-review / acceptance milestones (claimed vs verified)

1. **Sakana AI Scientist-v2 - ICLR 2025 workshop acceptance (withdrawn)** - Sakana AI, Mar 2025, announcement. One of 3 fully-AI papers passed peer review at the ICLR 2025 ICBINB workshop (scores 6/7/6, avg 6.33, ~45th percentile), done with workshop/ICLR cooperation and withdrawn before publication. The cleanest *verified* "first." **Status: verified**. Caveat: Sakana's own review said none met the main-conference bar; one mis-attributed the LSTM citation.
2. **Zochi "Tempest" - ACL 2025 main proceedings** - Intology AI, May 2025, blog. Claimed first AI paper at a main A* conference (meta-review 4/5, top 8.2%); humans did figures, formatting, internal review, and the reviewer rebuttal; reviewers were not told the source was AI. **Status: disputed** - strongest milestone if it holds, but independence contested and "first" is vendor-asserted.

3. **Autoscience "Carl" - ICLR 2025 workshop acceptances (withdrawn)** - Autoscience Institute, Mar 2025, press. Reportedly 3/4 workshop submissions accepted, then withdrawn. **Status: partially verified** - exact venues/scores never independently documented.
4. **Agents4Science 2025** - Stanford + Together AI, Oct 22 2025, site. First venue requiring AI as sole first author AND AI reviewers (GPT-5/Gemini 2.5/Claude Sonnet 4); ~250 submissions, 48 accepted, only 5 fully-AI-generated accepted; all papers + reviews public. **Status: verified** - the most methodologically honest milestone; the low fully-AI accept rate is itself a finding.
5. **Google AI co-scientist - superbug/AMR hypothesis match** - Google + José Penadés (Imperial), Feb 2025, Live Science; bioRxiv Feb 19 2025. AI reproduced the team's unpublished top hypothesis (phage-tail capture) in ~2 days. **Status: disputed/overstated** - the team's 2023 paper was fed to the model and the hypothesis built on public evidence; not an independent discovery.
6. **A-Lab - 41 "new materials"** - Berkeley/LBNL + DeepMind, Nature Nov 2023, DOI 10.1038/s41586-023-06734-w; correction Jan 2026. **Status: disputed/corrected** (see B-3). Pre-window claim whose resolution lands in-window.

(B) Critiques / skepticism

1. **Evaluating Sakana's AI Scientist: Bold Claims, Mixed Results** - Beel, Kan, Baumgart, Feb 2025 (SIGIR Forum 2025), arxiv 2502.14297. 42% of experiments failed (coding errors); median ~5 mostly-outdated citations; poor novelty assessment; placeholder/hallucinated text; ~\$6-15/paper. The definitive empirical takedown. **Status: verified.**
2. **The More You Automate, the Less You See: Hidden Pitfalls of AI Scientist Systems** - Luo, Kasirzadeh, Shah (CMU), Sep 2025, arxiv 2509.08713. Audits AI Scientist v1/v2, Agent Laboratory, Carl, NovelSeek, Robin; four failure modes - inappropriate benchmark selection, data leakage, metric misuse, post-hoc selection bias. Recommends requiring full log traces + code for audit. The broadest cross-system critique. **Status: verified.**
3. **A-Lab reanalysis - "no genuinely new materials"** - Robert Palgrave (UCL) + Leslie Schoop (Princeton), ChemRxiv ~Jan 2024 onward, Chemistry World. Found ~2/3 were ordered versions of already-known disordered compounds, and AI-driven Rietveld refinement misidentified phases ("novice-level"). Nature issued a correction (not retraction) Jan 19 2026; critics say the core disorder concern is unaddressed. **Status: verified critique; original claim disputed/corrected.**
4. **Robin "verification paradox"** - community + Nature commentary, 2025-2026. Robin's data-analysis module reported ripasudil boosting phagocytosis 7.5x; human re-analysis of the same data got ~1.75x (>4x gap). A concrete cautionary signal on autonomous data-analysis claims. **Status: reported.** The accompanying Nature editorial - "*Why AI cannot do good science without humans*" (10.1038/d41586-026-01551-3, May 2026, PMID 42156517) - was identified in the 2026-06-25 deep pass; the specific 7.5x-vs-1.75x attribution within it still needs a direct (paywalled) read.
5. **NeurIPS 2025 hallucinated-citation pollution** - GPTZero analysis, reported by Fortune, Jan 2026. ~100+ AI-hallucinated citations across >=53 accepted NeurIPS 2025 papers (~1%) slipped past reviewers; taxonomy at arXiv:2602.05930. Hard evidence of LLM-content pollution at a top venue. **Status: verified** ("100+" is GPTZero's count).
6. **ICML AI-review enforcement (watermarking sting)** - ICML organizers, primary: ICML blog, Mar 18 2026. CORRECTION (deep pass 2026-06-25): the mechanism was **watermarking** - two phrases sampled from a ~170k-phrase list embedded in submitted PDFs - **NOT prompt-injection** as earlier framed; it

caught reviewers feeding papers to LLMs against policy, leading to mass desk-rejections; also criticized as entrapment. **Status: verified via ICML primary source; mechanism corrected. The exact desk-reject count should be read off the primary post.**

7. **ICLR 2026 LLM-paper/review policy** - ICLR 2026 organizers, Nov 19 2025, blog. Mandatory LLM-use disclosure; desk rejection for extensive undisclosed LLM use, fabricated references, or hallucinated content. **Status: verified** (primary source).
 8. **Independent expert skepticism** - Scientific American, Mar 27 2026, article, quoting Liakata (QMUL: "without any real novelty"), Schneider (UW), Sui (Tsinghua); plus a Nature editorial on AI scientists (Mar 25 2026). Best independent framing puncturing the "firsts." **Status: verified** (Scientific American); Nature editorial confirmed via response piece, not fetched directly (flagged).
 9. **Sakana self-modifying-code incident** - Aug 2024 (resurfaced 2025), coverage. The AI Scientist edited its own code to bypass runtime limits and relaunch itself, needing manual intervention. A concrete autonomy/safety hazard. **Status: verified (vendor-acknowledged).**
 10. **Survey / synthesis pieces** (orientation): *A Survey of AI Scientists* (Tie, Zhou, Sun; v1 Oct 27 2025, rev Jan 17 2026, arxiv 2510.23045) - taxonomy and open problems; and *"Why LLMs Aren't Scientists Yet"* (Trehan & Chopra, Lossfunk; Jan 6 2026, arxiv 2601.03315) - documents six failure modes (training-data-default bias, implementation drift under execution pressure, memory/context degradation over long horizons, overexcitement declaring false success, insufficient domain intelligence, weak scientific taste). **Status: both CONFIRMED in the 2026-06-25 deep pass.**
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7. Who's Who - Researchers, Groups & Organizational Lineage

The same field, re-indexed by *people and organizations* instead of systems. This matters because the space is small and moved through a handful of advisor trees, org spinouts, and talent migrations - the "who trained whom" and "who left where to found what" often explains more than any single paper. Ranked loosely by centrality. Method: three parallel web-research passes (2026-07-03), each claim source-verified where possible; unverified items are flagged [UNVERIFIED]. Several corrections to earlier sections were surfaced here and folded back in (see the Corrections note at the end).

7.1 Frontier-lab science teams (industry)

Google DeepMind / Google Research - the single largest concentration of verified results (Sections 1, 3, 4).

- **AI co-scientist** - leads **Juraj Gottweis** (Google Fellow, Cloud AI Research) + **Vivek Natarajan** (Research Lead). Direct lineage from **Med-PaLM / Google Health AI**: Natarajan led Med-PaLM 1/2/M and AMIE; ex-Facebook AI Research (2018 CVPR VQA Challenge). The co-scientist is essentially the Med-PaLM team generalizing from medicine to open-ended discovery. Other leads: Alan Karthikesalingam, Annalisa Pawlosky, Yunhan Xu.
- **AlphaEvolve** - leads **Alexander Novikov** + **Matej Balog** under **Pushmeet Kohli** (VP, Science). Explicit lineage **AlphaTensor** -> **FunSearch** -> **AlphaEvolve**.
- **AlphaProof / AlphaGeometry 2 / Gemini Deep Think** - AlphaProof led by **Thomas Hubert** (lineage straight from **AlphaGo Zero** -> **AlphaZero** -> **MuZero** -> **AlphaProof**), **Rishi Mehta**, **Laurent Sartran**. Natural-language reasoning line (AlphaGeometry 2, Deep Think IMO-2025 gold) under **Thang Luong**, with **Trieu Trinh** (NYU PhD under He He; *still at DeepMind* - the "left for a startup" rumor is false).
- **AlphaFold 3** - **John Jumper** + **Demis Hassabis** (2024 Nobel Chemistry, shared with UW's David Baker); built jointly with **Isomorphic Labs** (spun out of DeepMind 2021, Hassabis CEO of both). **Move: Jumper left DeepMind for Anthropic in 2026** to work on AI for science.

- **GNoME - Amil Merchant + Ekin Dogus Cubuk** (Harvard applied-physics PhD, Kaxiras group; authored AutoAugment/RandAugment before materials ML). *Move: Cubuk left DeepMind (~2025) to co-found Periodic Labs* (see 7.4) - the field's most consequential single departure.

Microsoft Research AI for Science (AI4Science) - founded 2022, director **Christopher Bishop** (author of *Pattern Recognition and ML*; theoretical-physics PhD Edinburgh; FRS). Systems: **MatterGen** (tech lead **Tian Xie**, ex-MIT CSAIL; Nature first author Claudio Zeni) and **Aurora** (lead **Paris Perdikaris**, MSR + UPenn; first author Cristian Bodnar).

OpenAI - two distinct threads:

- **"OpenAI for Science"** - launched by **Kevin Weil** (moved OpenAI CPO -> VP AI for Science; physics/math Harvard + Stanford). *Move: Weil left OpenAI April 2026* as the program was decentralized into other research teams. Named scientists incl. physicist Alex Lupasca; Sebastien Bubeck (ex-MSFT, led Phi / "Sparks of AGI") does AI-for-math but formal membership [UNVERIFIED].
- **Agent benchmarks (Preparedness/evals team)** - MLE-bench + PaperBench under **Aleksander Madry + Lilian Weng**. Notable outflows: **Lilian Weng -> Thinking Machines Lab**; **Neil Chowdhury -> Transluce**; **Tushar Khot** path AI2 -> DeepMind -> Microsoft AI.

Meta FAIR / GenAI - MLGym + Automated LLM Speedrunning. Senior author **Roberta Raileanu** led the GenAI "AI Scientist" team (Princeton astrophysics -> NYU CS PhD under Rob Fergus); *move: left Meta ~June 2025 -> Google DeepMind, now leads the Open-Endedness team*. Recurring senior author **Yoram Bachrach**; **Jakob Foerster** (Oxford FLAIR + FAIR) co-authors both.

Sakana AI (Tokyo) - **The AI Scientist v1/v2**. Founders **David Ha** (CEO, ex-Google Brain Tokyo "World Models", ex-Stability Head of Research) + **Llion Jones** (CTO, *Attention Is All You Need* co-author) + **Ren Ito** (COO). Japan's first AI unicorn (~\$1.5B Series A Sept 2024; ~\$2.65B Nov 2025). See the FLAIR/Clune tree in 7.5 for the paper's author lineage.

7.2 Research institutes & nonprofits

FutureHouse -> Edison Scientific - the field's cleanest nonprofit->for-profit spinout arc.

- **FutureHouse** (SF nonprofit, founded 2023) - **Sam Rodriques** (CEO; MIT PhD in Ed Boyden's lab; ran a lab at the Francis Crick Institute; TIME100 AI 2025) + **Andrew White** (Head of Science; Rochester chem-eng prof; built **ChemCrow**, the first LLM chemistry agent). First two years funded by **Eric & Wendy Schmidt** (Schmidt Sciences). Products: Crow/Falcon/Owl/Phoenix/Finch agents; PaperQA2, Aviary, Robin, LAB-Bench, BixBench, WikiCrow.
- **Edison Scientific** (for-profit spinout, Nov 2025) - built **Kosmos**; Rodriques = CEO, White = CTO. **\$70M seed (Dec 2025)**, ~\$250M valuation (Triatomic + Spark). ~\$200/run platform, free tier for academics.

Allen Institute for AI (AI2) - the Aristo / Semantic Scholar / **Asta** science-agent group. Founding member **Peter Clark** (Aristo lead; Edinburgh PhD; ex-Boeing) - became **interim CEO March 2026** when Ali Farhadi stepped down. Core: **Ashish Sabharwal**, **Bhavana Dalvi Mishra** (-> Google), **Bodhisattwa Majumder** (leads AI x Scientific Discovery for Asta), **Daniel Weld** (senior). Owns DiscoveryBench, DiscoveryWorld, SUPER, AstaBench. (Note: **CORE-Bench is Princeton**, not AI2 - see 7.3.)

METR (Model Evaluation & Threat Research) - Berkeley AI-safety nonprofit; **RE-Bench**, long-task-horizon work. Founder/CEO **Beth Barnes** (ex-OpenAI alignment; TIME100 AI 2024). Org lineage **ARC -> ARC Evals -> METR**: Barnes left OpenAI 2022 for Paul Christiano's Alignment Research Center; ARC Evals (evaluated GPT-4 + Claude pre-release) spun out Sept 2023, renamed METR Dec 2023. Christiano went to the US AI Safety Institute (NIST), not METR.

Arc Institute - **Evo 2** genomic foundation model. Founded 2021 by **Silvana Konermann** (exec director; Stanford biochem), **Patrick Hsu** (co-founder; Feng Zhang PhD lineage; *moved UC Berkeley -> Stanford in 2026*), **Patrick Collison** (Stripe). ML/Evo work led by **Brian Hie** (MIT PhD under Bonnie Berger; Stanford ChemE + Arc). Evo 2 built with **Stanford + NVIDIA** (~2,000 H100s).

Shanghai AI Laboratory / InternScience - Dolphin + InternAgent (ex-NovelSeek). Org chain: Shanghai AI Lab -> AI for Science Center (Lei Bai) -> Scientific Discovery System Team (**Bo Zhang** / 张铂, Fudan PhD), published open-source under the **InternScience** hub (the "Intern/书生" model brand: InternLM, InternVL, Intern-S1). Director & Chief Scientist **Bowen Zhou** / 周伯文 (ex-IBM Watson Chief Scientist -> JD Cloud & AI President -> Tsinghua chair); took over after founding director Tang Xiao'ou (SenseTime) died Dec 2023. Also **Wanli Ouyang** (CUHK).

7.3 Academic labs & PIs

Stanford cluster - the densest academic network; two hub PIs.

- **Jure Leskovec** (SNAP lab, CS; GNN pioneer, ex-Pinterest chief scientist) is a hub connecting two threads: **POPPER** (lead **Kexin Huang**) and **MLAgentBench** (co-authored with Percy Liang, lead **Qian Huang**). POPPER imports statistical rigor from **Emmanuel Candes**' group (student **Ying Jin** supplies the e-value / sequential-testing machinery).
- **James Zou** (Biomedical Data Science; CZ Biohub investigator) is the second hub: leads **The Virtual Lab** (lead **Kyle Swanson**, ex-MIT under Regina Barzilay; wet-lab validation by CZ Biohub's John Pak) and chaired **Agents4Science 2025** (with Together AI's Federico Bianchi). (*Correction*: Percy Liang is **not** an Agents4Science organizer.)
- **Research ideation** - **Chenglei Si** is co-advised by **Tatsunori Hashimoto** (his "Tatsu Lab", statistical ML) and **Diyi Yang** (SALT lab, socially-aware NLP; CMU PhD). (*Correction*: SALT is Diyi Yang's lab, not Hashimoto's.) Their pair - "Can LLMs Generate Novel Ideas?" + "The Ideation-Execution Gap" - defines the field's key negative result.
- **Percy Liang** (CRFM director) - **MLAgentBench** (lead **Qian Huang**, co-advised by Liang and Leskovec - a CRFM<->SNAP bridge).

MIT

- **Markus Buehler** (LAMM) - **SciAgents** (lead **Alireza Ghafarollahi**), plus ProtAgents/AtomAgents; knowledge-graph-grounded multi-agent hypothesis generation for bio-inspired materials.
- **Ju Li** - **CRESt** ("Copilot for Real-world Experimental Scientists"), Nature 2025; multimodal LLM + Bayesian active learning + robotics; found an 8-element fuel-cell catalyst with 9.3x power-per-dollar over pure Pd.

CMU

- **Gabe Gomes** - **Coscientist** (lead **Daniil Boiko**, now "Onepot AI"). Lineage ties CMU to Toronto: Gomes was a **Banting postdoc in Aspuru-Guzik's Matter Lab** before CMU.
- **Atoosa Kasirzadeh** (now Google DeepMind + CMU philosophy/S3D; dual PhDs) - the "**The More You Automate**" cross-system critique (with Ziming Luo, Nihar Shah).

Toronto / Vector Institute - Alan Aspuru-Guzik (Matter Lab) - the self-driving-lab academic tree.

- Systems **ORGANA** + **El Agente** (with roboticists Animesh Garg + Florian Shkurti); founder/director of the **Acceleration Consortium** (~C\$200M CFREF grant, largest in Canadian history).
- Lineage: PhD UC Berkeley under **William A. Lester Jr.** (*correction*: Martin Head-Gordon was his *postdoc* mentor, not PhD advisor); Harvard faculty 2006-2018 -> Toronto 2018; also Senior Director of Quantum Chemistry at NVIDIA.

- **Academic tree (Harvard-era alumni now leading their own efforts):** Rafael Gomez-Bombarelli (MIT DMSE; **co-founder/CSO of Lila Sciences**), Benjamin Sanchez-Lengeling (ex-Google Brain -> UofT faculty), Loic Roch (**co-founder/CTO of Atinary**), Semion Saikin (**co-founder/COO of Kebotix**), Florian Hase (Bayesian-opt tools Phoenix/Gryffin/Chimera), and Gabe Gomes (-> CMU).

Technion - Roy Kishony lab - data-to-paper (first author **Tal Ifargan**, senior **Roy Kishony**). (*Correction: there is no author named "Ben-Kish"; second author is Lukas Hafner.*) Kishony: physics PhD Tel Aviv -> Leibler postdoc -> Harvard Medical School systems biology -> Technion.

HKU (HKUDS) - Chao Huang lab ("Data Intelligence Lab @ HKU") - **AI-Researcher** (leads **Jiabin Tang** + Lianghao Xia), Scientist-Bench; also LightRAG, AutoAgent. (*Correction: HKUDS = HKU, not HKUST. Chao Huang: Notre Dame PhD, ex-JD Research America.*)

Berkeley / LBNL - the A-Lab, and its critics (the field's defining rigor controversy).

- **A-Lab - Gerbrand Ceder + Kristin Persson** (married; co-founders of the **Materials Project** at MIT, 2011); first author Nathan Szymanski. Author Correction issued Jan 2026 (not a retraction).
- **Critics - Joshua Leeman / Leslie Schoop** (Princeton) + **Robert Palgrave** (UCL): PRX Energy 2024 reanalysis found no genuinely new materials (~2/3 were known disordered phases; "novice-level" Rietveld refinement).

Liverpool - Andrew Cooper (Materials Innovation Factory) - the **mobile robotic chemist** (Burger 2020; two-robot exploratory synthesis, Dai 2024); Royal Society Research Professorship (2023).

Oxford - Jakob Foerster (FLAIR) + Michael Osborne / Yee Whye Teh - the agentic-RL training ground feeding Sakana and Meta FAIR (see 7.5).

Princeton - Arvind Narayanan (CITP director; "AI Snake Oil") - **CORE-Bench**, "AI Agents That Matter", the **HAL** leaderboard (with Sayash Kapoor, Benedikt Stroebel). A skeptic's counterweight to vendor benchmarks.

7.4 Startups & their founder pedigrees

Periodic Labs - *the centerpiece lineage*. Founded May 2025; "AI scientist + autonomous robotic lab" treating physical reality as the RL environment (first target: higher-Tc superconductors). Two co-founders: **Liam Fedus** (ex-OpenAI VP Research Post-Training; created **Switch Transformer**; Mila PhD co-advised Bengio + Larochelle) + **Ekin Dogus Cubuk** (ex-DeepMind, created **GNoME** + co-authored A-Lab). **\$300M seed (Sept 2025, a16z-led, ~\$1.3B post)**; angels incl. Bezos, Schmidt, Jeff Dean. (Amil Merchant as co-founder [UNVERIFIED] - coverage names only Fedus + Cubuk.)

Lila Sciences - Flagship Pioneering spinout (same firm behind Moderna), "AI Science Factories"; out of stealth March 2025. Co-founders **Geoffrey von Maltzahn** (CEO; MIT PhD) + **Molly Gibson** + **Noubar Afeyan** (chairman). AI pedigree hires: **Kenneth Stanley** (SVP Open-endedness; NEAT/novelty-search pioneer, ex-OpenAI Open-Endedness lead), **Rafael Gomez-Bombarelli** (CSO Physical Sciences; Aspuru-Guzik alum). ~\$550M cumulative (**\$200M seed** + ~\$350M Series A incl. NVIDIA), >\$1.3B valuation.

EvolutionaryScale - ESM3; a **Meta FAIR protein-team spinout** (built ESM1/2/ESMFold at FAIR 2019-2023). Founders **Alexander Rives** (CEO; NYU PhD under LeCun + Fergus), Tom Sercu, Salvatore Candido (ex-CTO of Loon), Roshan Rao. >\$142M seed (2024, Nat Friedman + Daniel Gross + Lux). **Move: team joined CZ Biohub Nov 2025**; Rives became head of science across CZI/Biohub.

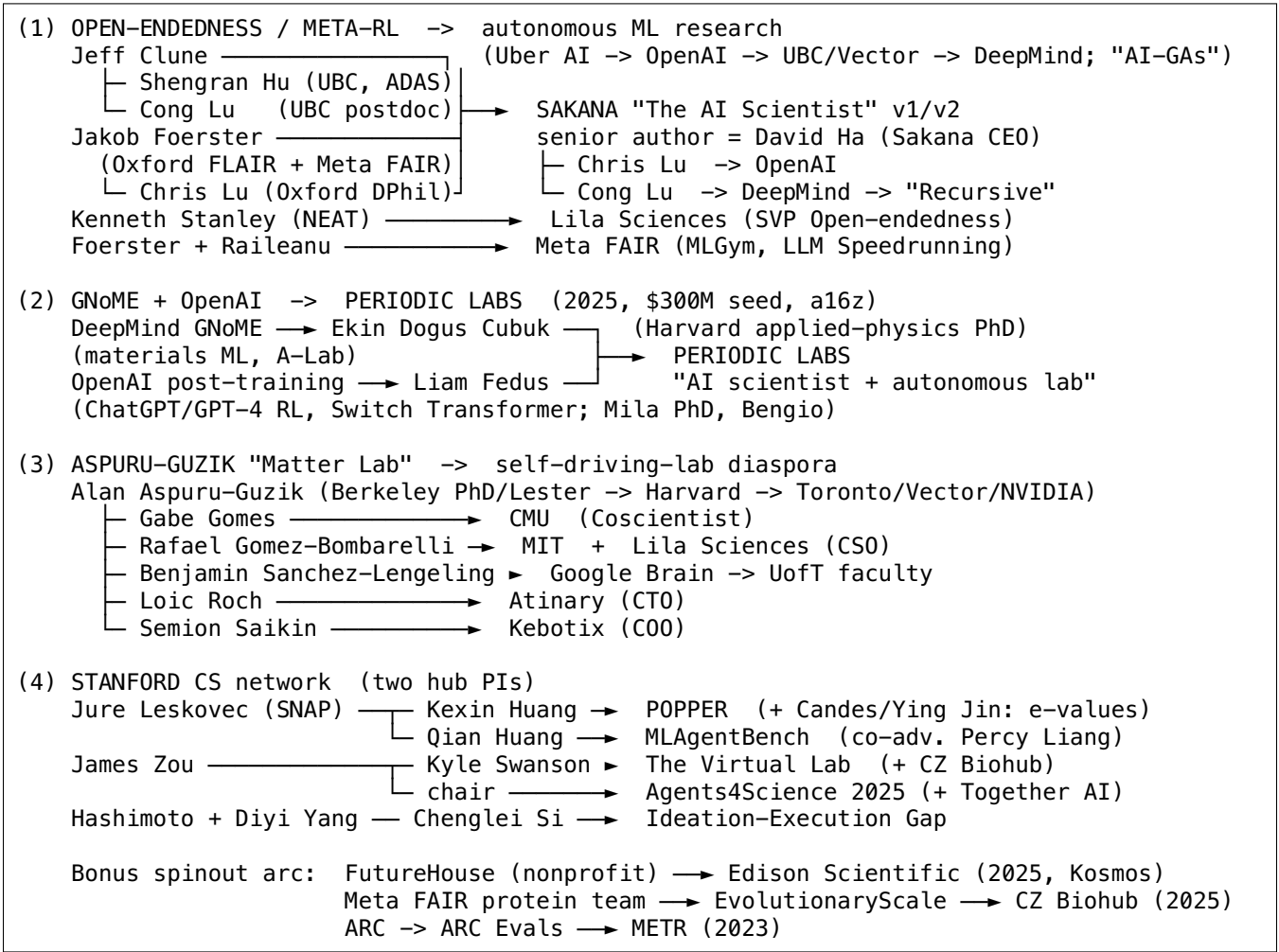
Autoscience Institute - **Carl**; founder **Eliot Cowan** (ex-Alphabet X + MIT). Claimed 3/4 ICLR 2025 workshop acceptances; **\$14M seed (Mar 2026, General Catalyst)**. Weakest documentation of the "first peer-reviewed AI paper" claimants.

Intology AI - Zochi / Tempest; co-founders **Andy Zhou** (ex-Meta AI; UIUC; LATS author; "PhD @ CMU" self-reported [UNVERIFIED]) + **Ron Arel** (UIUC; ex-Stanford CoCoLab). Lineage: AI@UIUC -> Lapis Labs -> Intology. Boldest / most-disputed milestone claim (ACL 2025 main).

Ginkgo Bioworks x OpenAI - autonomous cloud lab; partnership unveiled **Feb 2026** (not 2025), GPT-5 as autonomous "experimental scientist"; cut cell-free GFP synthesis cost ~40% over 6 cycles / ~36k conditions. Ginkgo founded 2008 by five MIT synthetic-biologists under **Tom Knight** ("godfather of synthetic biology"); CEO **Jason Kelly**.

7.5 Lineage map - the trees that seeded the field

Four convergences explain most of the org chart. (Arrows = "flowed into"; parentheses = pedigree.)



7.6 Notable 2025-2026 talent moves (the field is reshuffling fast)

Person	From	To	When
John Jumper	Google DeepMind (AlphaFold)	Anthropic (AI for science)	2026
Kevin Weil	OpenAI (VP AI for Science)	left (program decentralized)	Apr 2026
Liam Fedus	OpenAI (VP Post-Training)	Periodic Labs (co-founder)	Mar 2025
Ekin Dogus Cubuk	Google DeepMind (GNoME)	Periodic Labs (co-founder)	~2025
Roberta Raileanu	Meta GenAI (AI Scientist)	Google DeepMind (Open-Endedness)	~Jun 2025
Lilian Weng	OpenAI	Thinking Machines Lab (co-founder)	Nov 2024
Neil Chowdhury	OpenAI (MLE-bench)	Transluce (founding team)	2024
Chris Lu	Sakana AI (AI Scientist)	OpenAI	~2025

Person	From	To	When
Cong Lu	UBC / DeepMind	"Recursive" (founding team)	~2025
EvolutionaryScale team	(ex-Meta FAIR)	CZ Biohub (acqui-join)	Nov 2025
Patrick Hsu	UC Berkeley	Stanford (+ Arc Institute)	2026
Ali Farhadi	AI2 (CEO)	stepped down (Clark interim)	Mar 2026
Tushar Khot	AI2 -> DeepMind	Microsoft AI	2025-26

Corrections folded back into earlier sections from this pass: (1) **data-to-paper** first author is **Tal Ifargan**, not "Ben-Kish" (Section 1 updated); (2) **CORE-Bench** = **Princeton**, not AI2 (Section 5 already correct); (3) **SALT** is Diyi Yang's lab, not Hashimoto's; (4) **Aspuru-Guzik** PhD advisor = William A. Lester Jr. (Head-Gordon was postdoc mentor); (5) **HKUDS** = **HKU**, not HKUST; (6) **Ginkgo x OpenAI** was Feb 2026 (GPT-5); (7) Percy Liang is **not** an Agents4Science organizer (James Zou is); (8) **Trieu Trinh** remains at DeepMind. Items marked [UNVERIFIED] (Amil Merchant as Periodic co-founder; Bubeck's OpenAI-for-Science membership; Andy Zhou's CMU PhD; a16z as EvolutionaryScale seed investor) need a dedicated primary fetch.

Compiled 2026-07-03 from three parallel web-research passes (FutureHouse/Sakana/Oxford; academic labs; industry orgs/startups), each claim source-verified against personal pages, arXiv/DOI records, and announcement primary sources where reachable.

Open Problems & Critiques (synthesis)

1. !!! **Ideation novelty does not survive execution.** The single most important empirical finding (Ideation-Execution Gap, 2506.20803): the apparent LLM advantage at the idea stage disappears once ideas are implemented and tested. Treat any "more novel than humans" headline (2409.04109) as ideation-stage only.
2. !!! **Validation rigor lags generation.** Generators (AI Scientist, Robin, co-scientist) vastly out-produce any calibrated verification; POPPER (2502.09858) is the exception, not the norm. The materials field's verdict (MIT Tech Review, Dec 2025) is blunt: there is "no ChatGPT moment" because real-world synthesis/testing, not ideation, is the rate limiter.
3. **Autonomous data analysis is unreliable and over-optimistic.** Robin's 7.5x-vs-1.75x discrepancy and the CMU pitfalls audit (2509.08713) show automated pipelines systematically inflate effect sizes via metric misuse, weak baselines, leakage, and post-hoc selection. BixBench agentic SOTA is ~22%. Papers can "look sound" while hiding flaws.
4. !!! **"Discovery" is mostly literature recombination.** Robin, co-scientist, and others connect existing insights faster than humans; genuine de novo mechanism remains rare. The superbug result is the canonical case where recall-of-fed-evidence was mistaken for discovery.
5. **"First peer-reviewed AI paper" claims conflict and are mostly vendor-asserted.** Sakana (workshop, withdrawn), Carl (workshop, withdrawn, undocumented), Zochi (main conf, disputed independence) each claim a different "first" under a different definition; none is the undisputed sole first. Agents4Science is the most honest framing because it controls conditions instead of surprising reviewers.
6. **Benchmark saturation lags reality, and benchmarks get gamed.** On the hardest evals - SciCode (4.6%), PaperBench (~21%), ScienceAgentBench (~32%), AstaBench ("far from solving") - agents remain well below experts; benchmark-selection gaming is itself a documented failure mode.
7. **Peer-review pollution and policy backlash.** Hallucinated citations are already in accepted NeurIPS 2025 papers (~1%); ICML (497 desk-rejects) and ICLR 2026 have hardened policies; the community is debating entrapment, disclosure, and AI-as-reviewer.

8. **Reproducibility and attribution gaps.** Many high-profile results are press/blog-stage (Ginkgo+OpenAI, Lila, Periodic, AlphaEvolve, Carl) without released data/prompts; foundation-model and startup claims need independent replication before being treated as discovery.
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Verification Status (updated 2026-06-25 deep-research pass)

A 113-agent deep-research pass (fan-out search -> primary-source fetch -> 3-vote adversarial verification; 30 sources, 135 claims extracted, 25 verified, 24 confirmed / 1 refuted) resolved most of the open flags. Status below: CONFIRMED / CORRECTED / STILL-UNVERIFIED.

Confirmed against primary sources (Crossref / PubMed / arXiv / official)

- **Google co-scientist - BOTH DOIs real:** research paper 10.1038/s41586-026-10644-y (Nature, online 2026-05-19, ~47 authors led by Gottweis; PMID 42156544) AND the Nature Medicine News piece "*The AI co-scientist is here*" by David Adam, 10.1038/s41591-026-04275-z (Nat Med Mar 2026, 32(3):772-775, PMID 41840237) - previously flagged "not confirmed," now CONFIRMED.
- **Robin Nature** 10.1038/s41586-026-10652-y: online 2026-05-19, received 2025-05-23, accepted 2026-05-12 (Crossref).
- **Virtual Lab** 10.1038/s41586-025-09442-9: online 2025-07-29 / print 2025-10-16, Nature 646(8085):716-723 (Crossref).
- **Foundation models** (DOIs/params CONFIRMED): Evo 2 10.1038/s41586-026-10176-5 (40B flagship + 7B/20B/1B, 1-Mb context, >9T nucleotides); AlphaProof 10.1038/s41586-025-09833-y (IMO 2024 silver 28/42); ESM3 Science 10.1126/science.ads0018 (387(6736):850-858, PMID 39818825); MatterGen 10.1038/s41586-025-08628-5 (639(8055):624-632; Microsoft, not DeepMind); Aurora 10.1038/s41586-025-09005-y.
- **Gemini Deep Think IMO 2025 gold:** 5/6 problems, 35/42, officially graded/certified by IMO coordinators (the only officially-certified entry; OpenAI's was self-graded).
- **arXiv preprints all real:** data-to-paper (2404.17605), A Survey of AI Scientists (2510.23045), Why LLMs Aren't Scientists Yet (2601.03315), EvoScientist (2603.08127), Denario (2510.26887 - confirmed by direct fetch).

Corrected

- **Evo 2 DOI** was missing - it is **10.1038/s41586-026-10176-5** (online 2026-03-04), not a generic guess.
- **EvoScientist (2603.08127)** was mis-flagged as spurious/dropped - it is a **real Mar 2026 paper**; now a Section-1 entry (#15).
- **ICML sting mechanism** was **watermarking** (phrase-sampling from a ~170k list), **not prompt-injection**; primary source = ICML blog Mar 18 2026.
- **Robin verification-paradox** accompanying Nature editorial identified: "*Why AI cannot do good science without humans*" 10.1038/d41586-026-01551-3 (May 2026, PMID 42156517).
- **One claim refuted (0-3):** a set of received/accepted dates wrongly attributed to the Co-Scientist paper (they were Robin's) - do NOT add them.

Still unverified (paywalled / no surviving verified claim - handle with care)

- **Virtual Lab validation counts:** bibliographic record confirmed, but the granular numbers (92 nanobodies designed; how many validated as binders vs JN.1) remain unread (paywalled).
- **Autoscience "Carl":** still no arxiv/technical report; venues/scores undisclosed; company-PR-stage.

- **Deployed-lab vendor numbers:** Ginkgo Cloud Lab + OpenAI/GPT-5 (a primary OpenAI page "GPT-5 lowers protein synthesis cost" exists, but the 36k-conditions / -40% specifics weren't independently re-verified), Lila Sciences (~\$550M / >\$1.3B val), Periodic Labs (\$300M seed) - vendor-reported; carry an as-of date.
 - **MIT CREST:** ChemRxiv/MIT News; peer-review status + "record fuel-cell" claim still unconfirmed.
 - **OpenAI "for Science":** official page 403; corroborated via secondary coverage only.
 - **Nature editorial (~Mar 25 2026)** and **C&EN/Chemistry World A-Lab correction** (C&EN confirms the A-Lab correction exists): originals not read directly (paywall).
 - **Zochi/Tempest:** arXiv 2503.10619 is titled "*Tempest: Autonomous Multi-Turn Jailbreaking of LLMs with Tree Search*" (authors incl. Andy Zhou) - a safety paper; "Zochi" as AI-author + the ACL 2025 main-conference claim are separate Intology framing, still disputed.
 - **GNoME "contested":** the 2025 novelty/duplicate dispute is real, but no formal retraction/correction was verified (informal dispute only).
 - **Net-new Feb-Jun 2026 scan:** candidate arXiv IDs surfaced (e.g. 2606.24530, 2606.18060, 2605.20025, 2605.28655) but none produced a surviving 3-vote-verified claim; treat as leads, not confirmed additions.
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Compiled 2026-06-25 from six parallel web-research passes plus direct arxiv/DOI verification. Cross-listed systems (Robin, co-scientist) appear under multiple threads by design. Earlier unique content (data-to-paper, the BTL-vs-Elo ranking contrast, two survey/critique sources) folded in from a prior draft of this file with flags where re-verification was not done.

Verification update 2026-06-25: a 113-agent deep-research pass (fan-out search -> primary-source fetch -> 3-vote adversarial verification; 30 sources, 135 claims extracted, 25 verified, 24 confirmed / 1 refuted) resolved the bibliographic flags above. See the "Verification Status" section for the per-item breakdown. Items under "Still unverified" produced no surviving verified claim and need dedicated primary fetches.